

sienna

Design

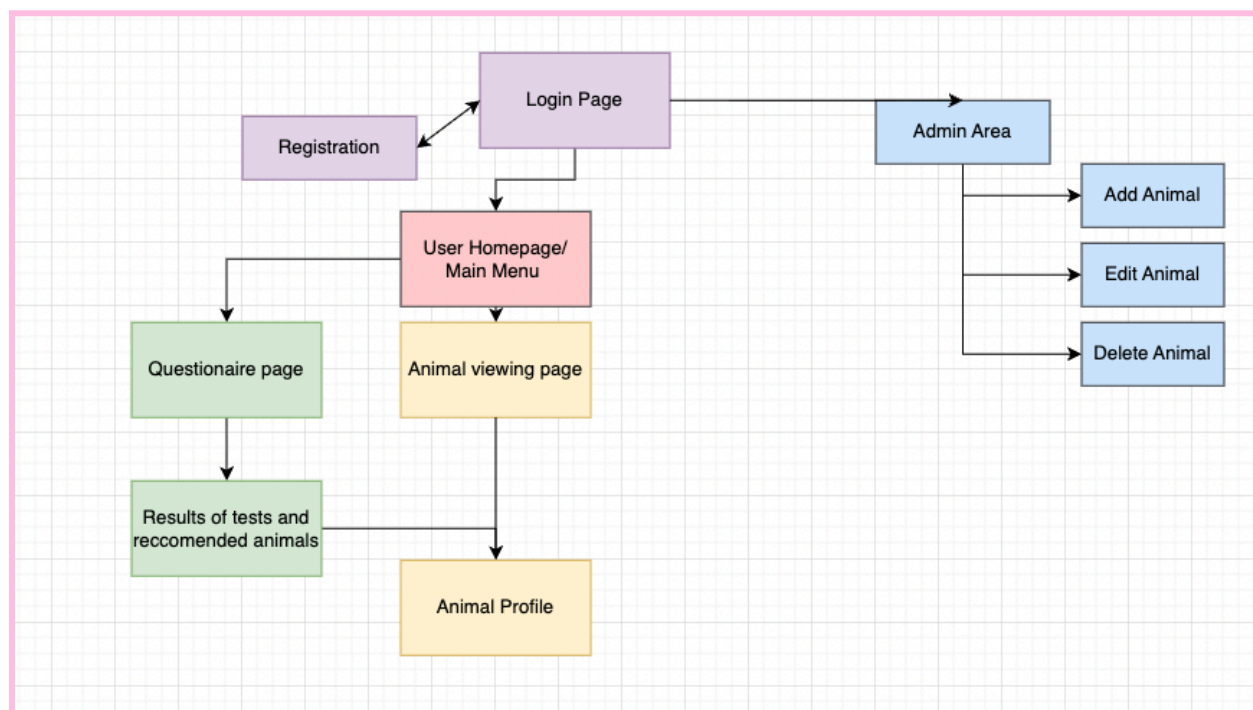
✓ Introduction/Overall

Proposed Layout

The system can be decomposed into five main components. Separating the system into smaller components makes it easier to design, develop, test and maintain whilst each section performs a specific function.

Section	Front End Justification	Backend Justification
User/Admin Authentication	Will allow users to register and existing users to login, and allow admins to access their page safely and securely. Registration will collect the users name, email, age and password. The interface should provide clear feedback when details are missing or forgotten from its validation system. It will also show an error message if login details are incorrect	The system will validate entered details, such as: - email cannot be blank and must be valid and unique - Password minimum 8 characters - Age must be a valid number - Login fields cannot be blank The passwords will be hashed in order to be kept secure and safe. During login, enter details will be compared against records in the users table. A session will then be created when authentication is successful. For the admin side, if the login details are incorrect it will not allow them access, which will stop unauthorised users from adding, editing or deleting animal records.
User Homepage/Main Menu	Will provide the main navigation area after the user logs in. It will allow the user to access the questionnaire, browse available animals and view their account. It will also explain the purpose of the adoption matching system. It will be clear and welcoming, and not overwhelming for the user.	The system will check the user's session to make sure they are logged in. It may retrieve the email and recent questionnaire information from the database so that personalised information can be displayed.
Administrator Menu	Provides staff with a dashboard to manage existing animals and their records. Admins will be clearly able to add new animals, edit existing details and remove records. To add animals it will store: Name, species, age, breed, sex, personality and exercise needs,	The system will check that the logged in account has the admin role before access is allowed. Parameterised SQL commands will insert, update and delete animal records. All inputs will be validated before changes are made to the database. A log

	children compatible, garden required, medical notes and an image. Forms will be clearly Labelled and display confirmation or error messages.	out function will clear the admins session and return them to the login page, preventing another person using the same device which will prevent unwanted access to the database.
Animal Viewing	Allows users to browse available animals using clearly presented animal cards. Each card will show information such as the animal name, image, species, breed and age. Users will be able to select a card to view a more detailed animal profile.	Animal records will be retrieved from the animals table. Search and filter selections will be used to change the database query or filter the retrieved records. Selecting an animal will use its unique animal ID to load the correct profile.
Questionnaire and Recommendations	Will use a form in order to allow users to answer questions about their home, lifestyle, experience and preferences. The questionnaire will have questions such as the housing type, access to garden, animal loneliness times, children, pets, experience, activity labels, what species and size preferred. These questions would not appear all at one time to not overwhelm the user and the input type would be drop downs or radio buttons. After submitting the questionnaire, the system will display compatibility scores almost immediately and recommended animals. It will also explain requirements the user does not meet.	Responses will be validated and compared with the requirements of each available animal. A weighted compatibility algorithm will calculate a percentage score for every animal. The animals will then be sorted from highest to lowest score so the top recommendations can be displayed. The questionnaire responses and match results may also be stored within the database with the users ID (email)



Database structure

Tables	Purpose
Users	Stores registered user and administrator account details
Animals	Stores user profiles, care needs and matching requirements
Questionnaire responses	Stores answers submitted by users
Matchresults	Stores calculated compatibility scores between responses and animals

One user can complete multiple questionnaire responses, each questionnaire can produce multiple compatibility results because it is compared to every available animal. Each animal can therefore appear in multiple match results belonging to different users or questionnaire attempts.

- Users **1** → **Many** Questionnaire responses
- Questionnaire responses **1** → **Many** matchresults
- animals **1** → **Many** matchresults

Validation

All user entered data should be validated before it is processed or stored in the SQLite database. Streamlit controls such as dropdowns radio buttons and number inputs will restrict the range of values users can enter. However, the backend will still validate the submitted values because interface controls alone should not be relied on.

Login + Administration

Login and Registration

User/Admin Authentication	Will allow users to register and existing users to login, and allow admins to access their page safely and securely. Registration will collect the users name, email, age, address and password. The interface should provide clear feedback when details are missing or forgotten from its validation system. It will also show an error message if login details are incorrect	The system will validate entered details, such as: - Username cannot be blank - Password minimum 8 characters - Required questions cannot be skipped - Age must be a valid number - Duplicate usernames not allowed The passwords will be hashed in order to be kept secure and safe. During login, enter details will be compared against records in the users table. A session will then be created when authentication is successful. For the admin side, if the login details are incorrect it will not allow them access, which will stop unauthorised users from adding, editing or deleting animal records.
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Interface Design

[insert cute emoji]

Company name

Find a companion which fits your life.

[some writing about the website]

Welcome Back!

Login to continue

Email

Enter your email

Password

Enter your password

Log in

Don't have an account? [Register here](#)

Branding and colour - A limited pink, white and dark grey palette will create a recognisable identity without being too distracting and bright. The system name will confirm that the user is on the right adoption website, and the dark text will keep information readable.

Mini introduction - the short statement will explain a bit about the system, this will give context before login and will reassure the first time users about the purpose of the website

Two columns - The branding is separate from the login. This will allow the user to focus on one point at a time which will prevent them from becoming overwhelmed increases usability.

Clear button - the large pink login button shows the users where to press. It is the only emphasised action so it will help users complete the intended task and reduce misclick (instead of pressing register here)

Clear route for new users - registration is shown as a secondary link below the form because it is an alternative task. It's easy for a new user to find without it looking larger than login

Whitespace and alignment - all the writing and textboxes are consistently aligned and have clear spacing. This separates sections and will lowers the risk of the user selecting wrong section. The amount white space reduces chance of user becomes overwhelmed.

Clear visuals - Large bold heading will welcome users and clearly identifies the page. It is followed by the task instruction. The difference in size and position will guide the user to page purpose to form clearly.

Boxing inputs - the email, password and the login buttons are all within one section which shows they are all together very clear for the user making the task readable.

Labels and prompts - The email and password boxes are labelled with placeholder text to show the user where the information should be entered. Double labels will ensure the user will know where to enter data clearly.

Validation and feedback - blank or invalid inputs will produce a specific message beside relevant field and a red error state. Password characters will be masked and failed login feedback will not reveal whether account exists.

Overall, these usability features will follow my client interview 11 and 12, stakeholder 6, competitors 1 (due to the colour combination), 2 (the formatting), 3(the headings and and clear sections)

[insert cute emoji]

Company name

Find a companion which fits your life.

[some writing about the website]

Get started

Create your account

Complete each field below, all fields required.

Full Name

Email

Password

Confirm Password

Age

Create Account

Already have an account? [Login](#)

Branding and colour - A limited pink, white and dark grey palette will create a recognisable identity without being too distracting and bright. The system name will confirm that the user is on the right adoption website, and the dark text will keep information readable. (Same as Login)

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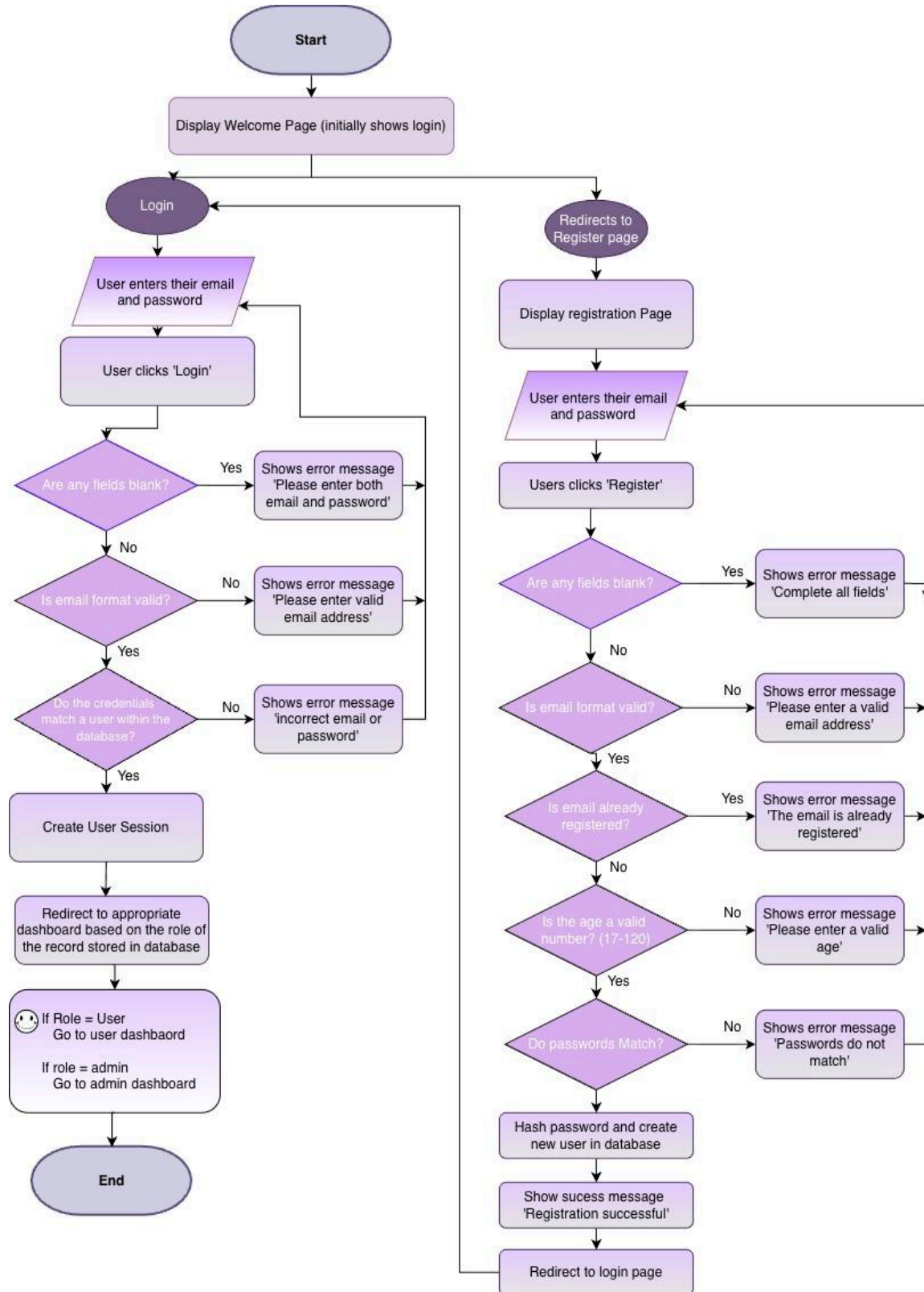
Age

Create Account

Already have an account? [Login](#)

Same annotations as LOGIN

Algorithm (Flow chart, pseudocode)



Pseudocode

Database (Table structure)

Field	Data type	Key/validation
user_id	INTEGER	Primary key which will auto increment
name	TEXT	required
email	TEXT	Required and unique
age	INTEGER	required , valid range
password_hash	TEXT	required
user_role	TEXT	'User' or 'admin'
date_created	DATE	required

Variables

Variable Name	Data Type	Purpose	Justification	E.g or Validation
user_name	String	Stores the name entered during registration	This will allow the system to identify and personalise the account. A string would be required for the datatype as it will contain	Cannot be blank

			alphabetic characters and may include spaces and hyphens.	
email	String	Will store users email address	It will be used as a unique identifier for logging in and contacting the user. It will contain letters and symbols so will be stored as a string.	Must follow a valid email format and be unique
age	integer	Stores the users age	It will be used to ensure the user meets any minimum age requirements. An integer is suitable for this variable as age is represented by a whole number.	Must be a valid positive number.
password	string	Temporarily stores the password entered by the password	It is needed to authenticate the user securely. Passwords usually contain letters, numbers and special characters, so a string is suitable for the data type.	Minimum of eight characters.
confirm_password	string	Stores the repeated password entered during registration	This allows the system to check that the user has mistyped their password before creating the account.	It will have to match password
password_hash	string	Stores the hashed version of the password	Passwords should not be stored as plain text. Storing the hash reduces the risk of passwords being exposed if the database is accessed without permission.	Generated using a secure hashing function
login_email	string	Stores the email entered on the login page	The variable is required so that the system can search for the corresponding account in the database	Cannot be blank
login_password	string	Temporarily stores the password entered during login	The entered password will be checked against the stored password hash in order to verify users identity	Cannot be blank
user_id	integer	Uniquely	An integer is efficient for use as a	It will be retrieved

		identifies the logged in user	database primary key, it will allow records such as questionnaires and recommendations to be linked to the correct user.	from the database after login
user_role	String	Stores whether the account is an administrator or a user	This variable is required for access control, so that normal users cannot access the administrator pages Controls access to the administrator area, although normal registrations should automatically receive the role 'user', the admin role should not be selectable from the registration form.	Will be stored as 'user' or 'admin'
error_message	string	Stores the error message which will appear when the input or login details are invalid	My system will provide clear feedback and help the user understand what has gone wrong and how to correct it. A string will be suitable for this message as it contains many words in one sentence.	'Incorrect email or password, please retry'
is_authenticated	boolean	Records whether login attempt was successful	A boolean data type is suitable as authentication only has two possible values.	True or False

Validation

1. Name

- Presence check: Field cannot be empty or contain only spaces
- Length check: name must be between 2 and 50 characters
- Whitespace check: spaces at beginning and end will be removed

2. Email address

- Presence check: Field cannot be empty or contain only spaces
- Format check: value must contain a @ and domain
- Uniqueness check: a database query will check the address not already registered
- Whitespace check: spaces at beginning and end will be removed
- Check characters: convert all to lower case

3. Age

- Presence check: Field cannot be empty or contain only spaces
- Type check: data must be a whole number
- Range check: age must be 18+

4. Password

- Presence check: Field cannot be empty or contain only spaces
- Length check: password must be more than 8 characters

Registration

- Confirmation check: the confirmation field must exactly match original password
- Security processing: the valid password will be hashed before it is inserted into the database and the original password will never be stored.

Login

- The password will be checked against the stored password hash rather than compared as plain text.

Registration outcome

- The account will be created if every field passes validation, individual messages will be shown if one or more fields are invalid.

Session check

- A session will only be created after successful authentication. Invalid information will leave users on the login page and will not allow them to access the user or admin page.

💖 Admin Page

Admin Page

Administrator Menu	Provides staff with a dashboard to manage existing animals and their records. Admins will be clearly able to add new animals, edit existing details and remove records. To add animals it will store: Name, species, age, breed, sex, personality and exercise needs, children compatible, garden required, medical notes and an image. Forms will be clearly Labelled and display confirmation or error messages.	The system will check that the logged in account has the role before access is allowed. Parameterised SQL commands will insert, update and delete animal records. All inputs will be validated before changes are made to the database. Will also include a logout function, to securely log out so no physical access to the system unless its an admin.
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Interface

The image shows a web application for managing animal records. The layout includes a sidebar on the left and a main content area.

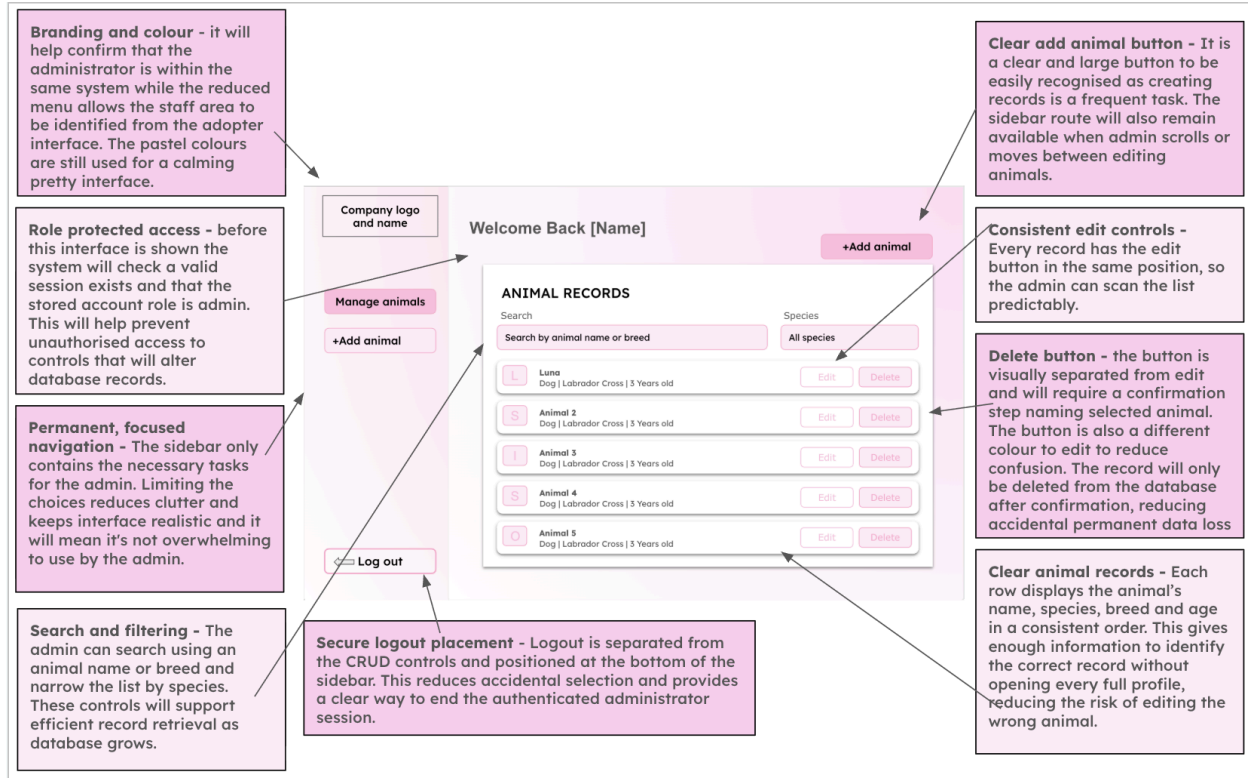
Sidebar:

- Company logo and name
- Manage animals
- +Add animal
- Log out

Main Content Area:

- Welcome Back [Name]
- +Add animal
- ANIMAL RECORDS**
- Search: Search by animal name or breed
- Species: All species
- Table of animal records:

Letter	Animal Name	Breed	Age	Edit	Delete
L	Luna	Dog Labrador Cross	3 Years old	Edit	Delete
S	Animal 2	Dog Labrador Cross	3 Years old	Edit	Delete
I	Animal 3	Dog Labrador Cross	3 Years old	Edit	Delete
S	Animal 4	Dog Labrador Cross	3 Years old	Edit	Delete
O	Animal 5	Dog Labrador Cross	3 Years old	Edit	Delete



Company logo and name

Manage animals

+Add animal

Log out

Add a new animal

ANIMAL INFORMATION

Animal name

E.g Luna

Age

E.g 3

Species

E.g Dog

Sex

E.g Male

Breed

E.g Labrador Cross

Image URL

E.g google.com/images

Description

Describe animals routine, habits, behaviour and ideal home

Matching requirements

Used by the compatibility algorithm

Save animal

Cancel

Consistent administrator layout - The same logo, sidebar position, colours are reused from manage animals. Consistency reduces the amount the administrator must learn and makes it clear that both pages belong to one CRUD workflow.

Validated save and safe cancellation- Save Animal is the strongest action and will run validation before a parameterised INSERT transaction. Success feedback is shown only after the record is committed to the database. Cancel returns to the list without writing partial data.

Numeric age validation - Age will use a numeric control and reject blank, negative or unrealistic values. The unit will be stated as years so the stored value has one consistent meaning for display, filtering and compatibility calculations.

Controlled selection values - Species and sex use drop-down controls rather than unrestricted text. Restricting entries to supported values prevents spelling variations and allows reliable database filtering and comparisons.

Validated image - The image field entered value will be checked before use. A valid direct image URL or uploaded file should be stored.

Guided multi-line description - The larger description area prompts the admin to record personality, routine, behaviour and ideal home information. A sensible length limit prevents empty or excessively long public profile.

Clear heading and grouped form - The page title states the current task, while all related controls are contained within one Animal Information card. Grouping the inputs shows that they will be validated and submitted as one database record.

Labels and examples - Labels remain after data is inputted and short examples are shown to show the expected format.

Required text-field validation - The information cannot be blank or contain only spaces. Inputs will be checked before insertion and a specific message will appear beside the affected field without removing other valid entries.

Secure exit from the admin session - Logout remains separate from Save and Cancel, preventing admins from being confused. Selecting it clears the active session before returning to the login interface.

Expandable matching requirements - Less frequently used algorithm fields are kept inside one expandable section so the initial form remains simple. It will collect size, activity level, home space, garden, children, pets, experience and maximum time alone.

Company logo and name

Manage animals

+Add animal

Logout

Add a new animal

ANIMAL INFORMATION

Animal name

Eg Luna

Age

Eg 3

Species

Eg Dog

Sex

Eg Male

Breed

Eg Labrador Cross

Image URL

Eg google.com/images

Description

Describe animals routine, habits, behaviour and ideal home

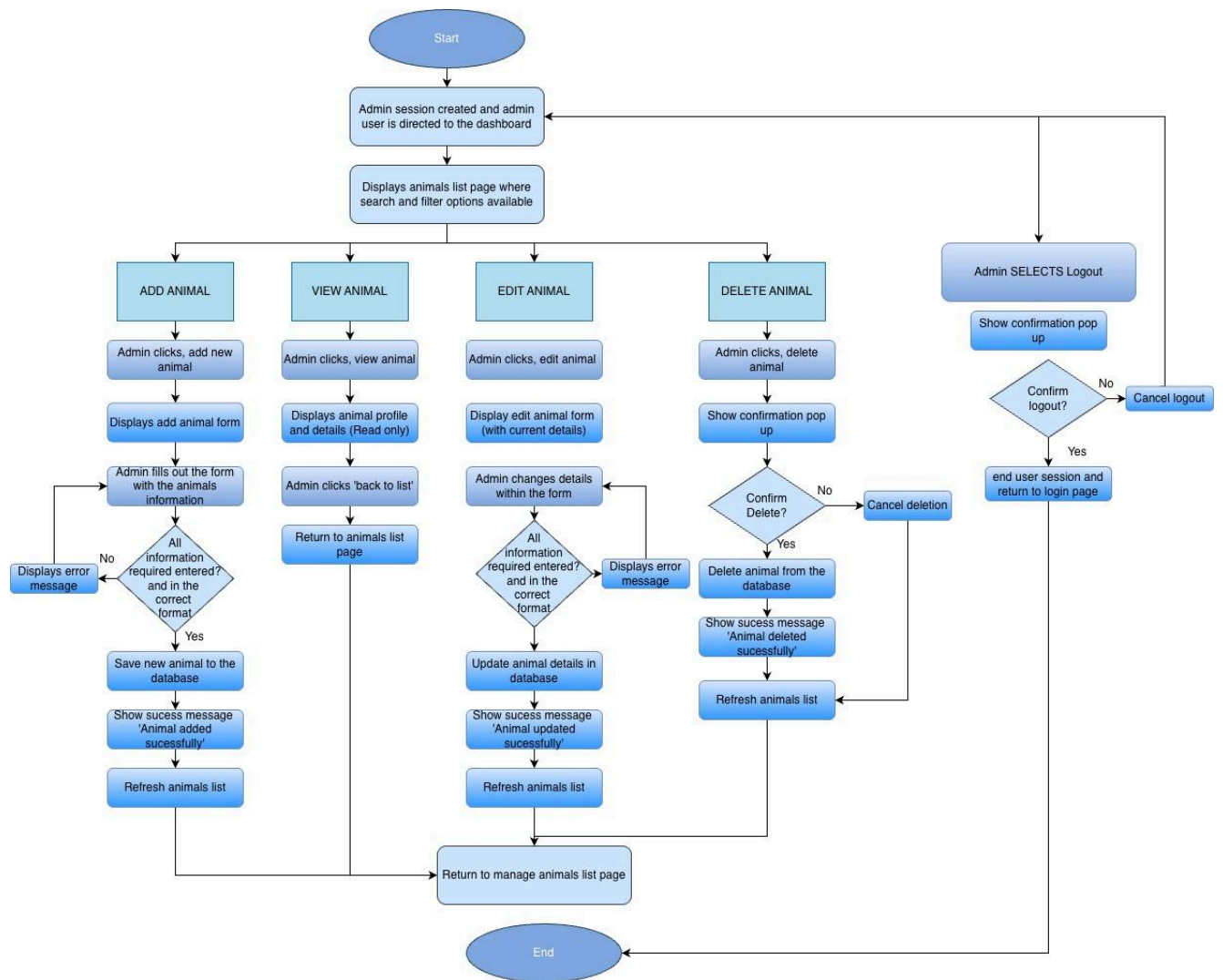
Matching requirements

Used by the compatibility algorithm

Save animal

Cancel

Pseudocode



Animal Table

Field	Data type	Key/validation
animal_id	INTEGER	Primary key, autoincrement
name	TEXT	required
species	TEXT	Required - dog, cat or rabbit

breed	TEXT	Required
age	INTEGER	Required, cannot be zero
sex	TEXT	required- male, female or unknown
size	TEXT	Required - small medium or large
personality	TEXT	required
Exercise_needs	INTEGER	Required - 1 to 3
minimum_housing_level	INTEGER	Required - 1 to 3
garden_required	BOOLEAN	Required 0 or 1
child_compatible	BOOLEAN	Required 0 or 1
max_hours_left_alone	INTEGER	required
Medical_notes	TEXT	optional
image	TEXT	Required, an image path
date_added	DATE	required

Variables

Variable Name	Data Type	Purpose	Justification	Validation or e.g
animal_id	INTEGER	Uniquely identifies	A unique integer	Automatically

		each animal record	stops confusion between animals with similar names and allows the correct to be updated and deleted	generated
name	string	Stores the animals name	The name is required to identify the animal throughout the interface and allows the user to remember the animal with a specific name. A string is suitable as it allows letters spaces and punctuation.	Can't be blank
species	string	Stores the type of animal	Species is required for searching, filtering and matching against user preferences.	dog, cat or rabbit
breed	string	Stores the animals breed	Breeds provides users with important information and may help describe likely size or care requirements	Can't be blank or might be 'unknown'
age	INTEGER	Stores the animals age	Age affects care needs. An integer is suitable as it allows the value to be validated sorted and compared	cannot be zero
sex	string	Stores the animals sex	This is relevant profile information and may be considered by the potential adopters	male, female
size	string	Stores the animals	Size affects space	small medium or large

		size and is used in filtering	requirements and is needed by the filtering and compatibility systems	
personality	string	Stores a description of the animals behaviour and personality	A text field will allow the admin to explain the animals personality and background in more detail to help attract possible adopters	Cant be blank but a character limit may be applied
Exercise_needs	INTEGER	Stores whether the animal has low, medium or high exercise needs	This can be compared with the adopters lifestyle to produce a more appropriate compatibility score. An integer is suitability for easier comparison	1 to 3 (low medium or high)
minimum_housing_level	INTEGER	Stores the amount of living space required	This can be compared with adopters lifestyle to aid the algorithm when creating the compatibility score	1 to 3
garden_required	boolean	Records whether the animal requires access to a garden	This may act as a compulsory suitability rule. A boolean value is useful as it is simple to input.	True or false
child_compatible	boolean	Records whether the animal is suitable for a household with children	There are only two options which make boolean values suitable. This is very important for the safety and welfare of the children and animals.	True or false

max_hours_left_alone	INTEGER	Stores value for which the animal can be left home alone for, and representing if more independent or clingy	This is important if a user works long hours and the animal requires a lot of attention it may cause neglect of the animal which is a key issue we are trying to prevent.	Can't be blank
Medical_notes	string/list	Stores any medical conditions, treatments or special requirements	Potential adopters need to understand any medical needs before making a decision. A string will allow multiple to be described.	May be blank or 'N/A'
image	string	Stores the location of the animal's image	This allows the correct photo to be displayed without storing the full image directly in database	Must use an accepted file extension
form_action	string	Stores the operation the admin is performing	The same form may be used for adding and editing variables. This variable will allow the program to select the right query.	'Add' or 'edit' or 'delete' or 'view'
delete_confirmation	boolean	Records whether the administrator has confirmed deletion	Deletion is permanent so confirmation reduces risk of animal being removed accidentally and not being considered for adoption	True or false
record_found	boolean	Records whether the database has retrieved the animal record successfully	This prevents the program from attempting to edit or delete a record not	True or false

			present in the database	
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Validation

1. Animal name

- Presence check: Cannot be empty
- Length check: Between 1 and 50 characters
- Format check: Letters, spaces, apostrophes, hyphens and appropriate numbers may be allowed

2. Species

- Presence check: Must be selected.
- Lookup check: Must be Dog, Cat or Rabbit.
- A drop-down menu will restrict the choices.

3. Breed

- Presence check: Cannot be empty
- Length check: For example, between 2 and 50 characters
- Format check: Must not consist entirely of unsupported symbols

4. Age

- Presence check: Must be provided
- Type check: Must be a whole number
- Range check: Cannot be negative and must be within the designed maximum animal age

5. Sex

- Lookup check: Must be Male, Female or Unknown
- A drop-down menu prevents invalid values

6. Size

- Lookup check: Must be Small, Medium or Large

7. Personality

- Presence check: A description must be provided
- Length check: For example, between 20 and 1,000 characters
- Whitespace check: A value consisting only of spaces will be rejected

8. Exercise need

- Presence check: An option must be selected
- Range check: Must be between 1 and 3

9. Minimum housing requirement

- Presence check: Must be selected
- Range check: Must be between 1 and 3

10. Maximum hours alone

- Type check: Must be an integer
- Range check: Must be between 0 and 24

11. Garden required

- Type check: Must be stored as a Boolean value 0 or 1

12. Child compatibility

- Type check: Must be stored as a Boolean value 0 or 1

13. Medical notes

- This field may be optional
- Type check: it will have a maximum length
- Text will be stored as plain text rather than interpreted as code

14. Animal image

- Presence check : An image must be uploaded if every profile requires one
- File-type check : Only approved formats such as JPG, JPEG and PNG will be accepted
- File-size check : The file must not exceed the chosen maximum size
- Filename handling : A safe, unique filename will be generated to avoid overwriting existing images

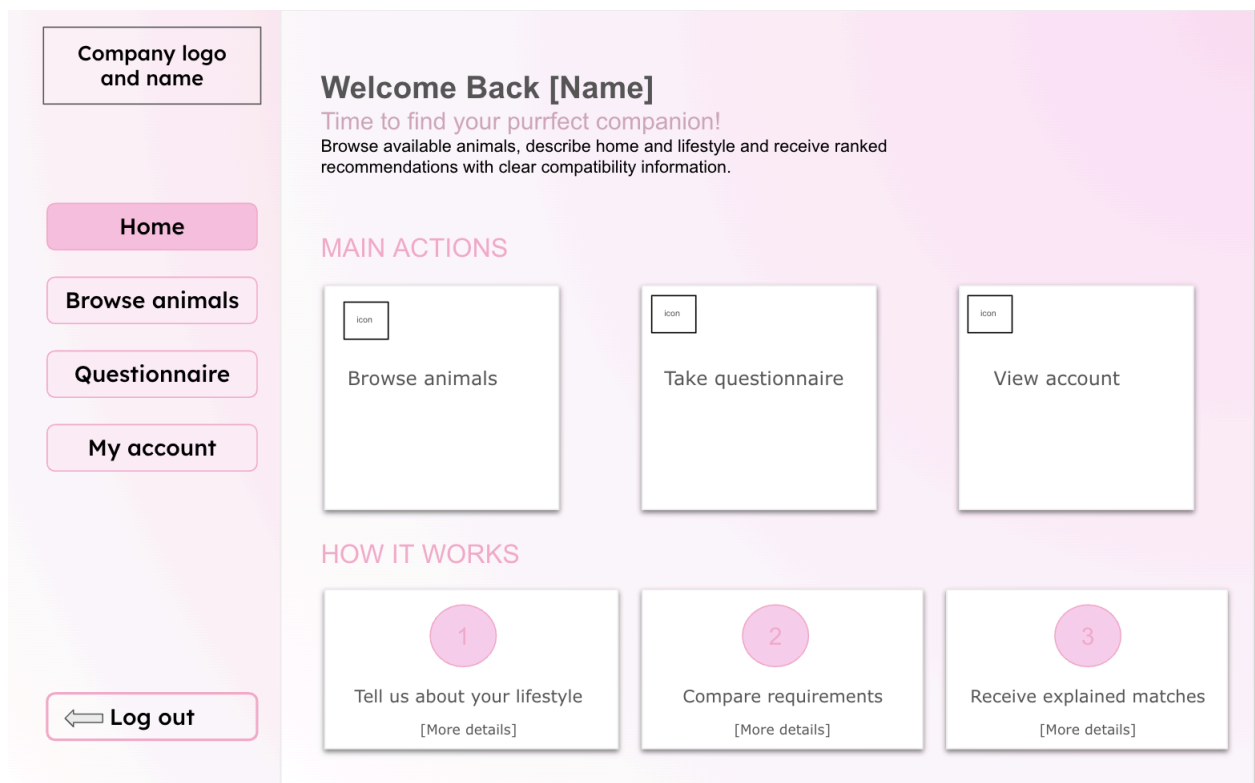
The animal record will only be inserted after all required fields pass validation. The SQL statement will be parameterised and committed only after successful validation. A success message will then confirm that the record has been created.

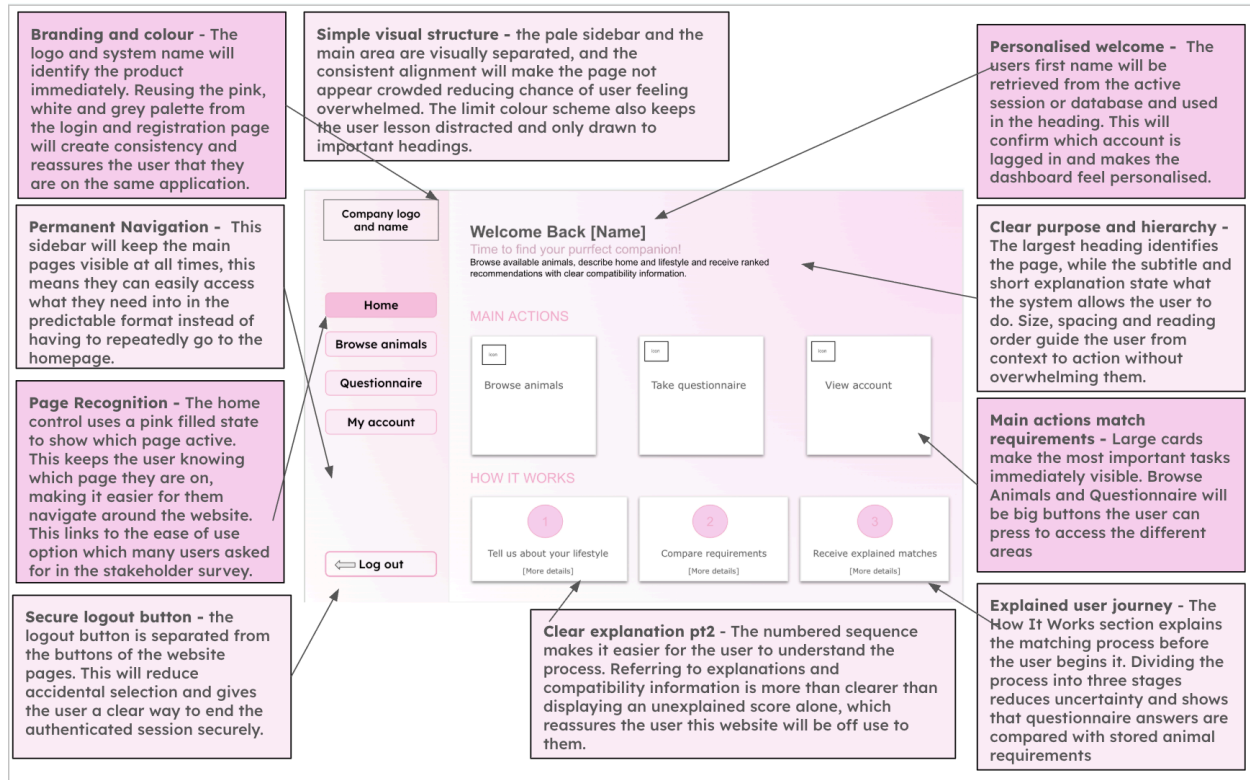
Main Menu/INterface

Main Menu

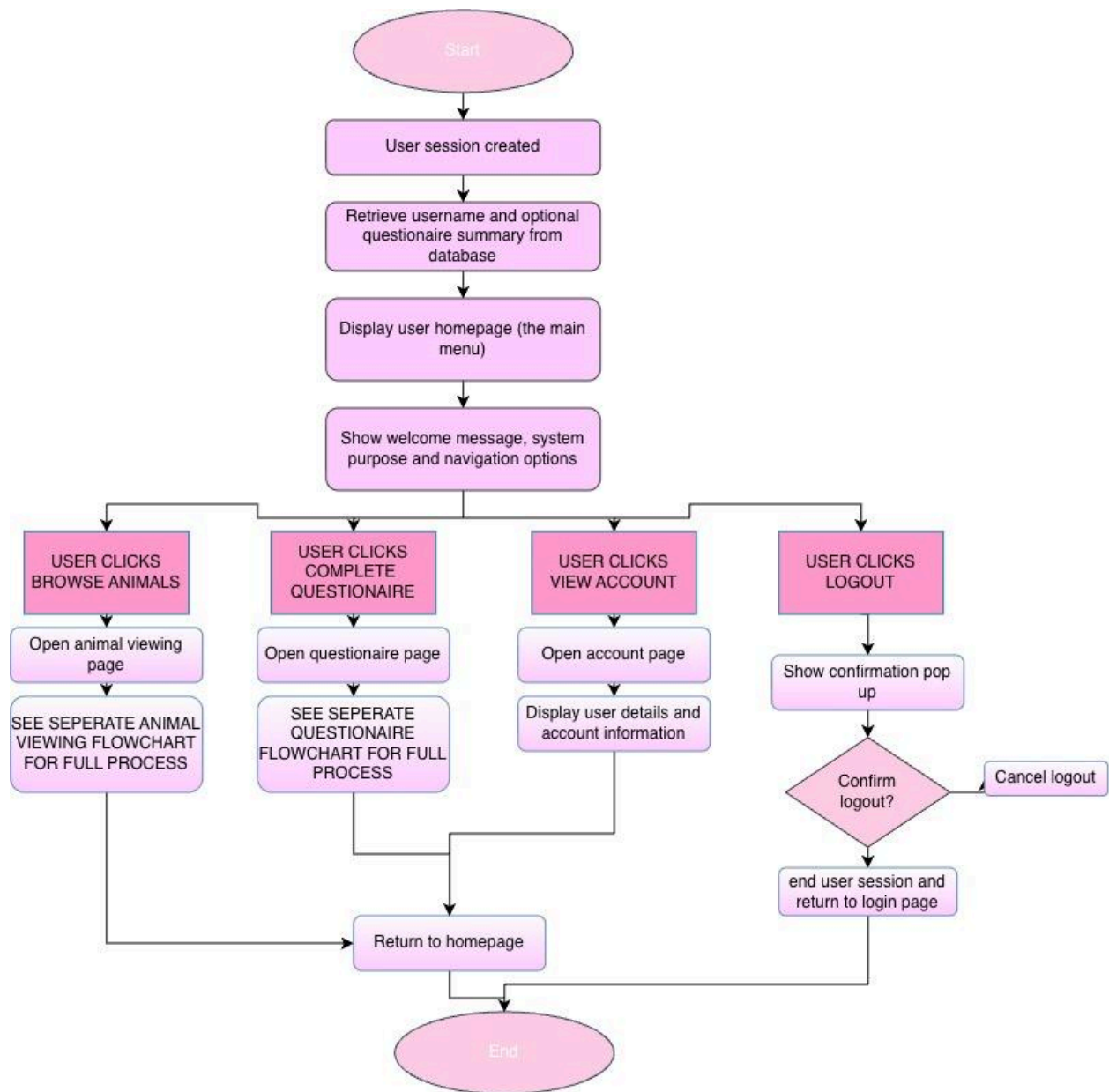
User Homepage/Main Menu	Will provide the main navigation area after the user logs in. It will allow the user to access the questionnaire, browse available animals and view their account. It will also explain the purpose of the adoption matching system. It will be clear and welcoming, and not overwhelming for the user.	The system will check the user's session to make sure they are logged in. It may retrieve the user's name and recent questionnaire information from the database so that personalised information can be displayed.
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Interface Design





Algorithm (Flow chart, pseudocode)



Pseudocode

Variables

Variable Name	Data Type	Purpose	Justification	E.g or validation
current_user_id	INTEGER	Identifies user currently accessing dashboard	Ensures that the dashboard displays information belonging to the correct account	Retrieved from session
current_user_name	STRING	Stores the logged in users name	This will allow the interface to display personalised information and reassures user they are accessing correct account	'Welcome sienna'
animal_count	INTEGER	Stores the number of animals currently available	It will provide a useful summary of information on the dashboard. An integer is a suitable data type as the number of animals is a whole number	Must be zero or larger
questionnaire_completed	BOOLEAN	Records whether the user has completed the questionnaire	This variable allows the system to display recommendations or prompt the user to complete the questionnaire. Boolean is suitable as there is only two possible states	True or False
recommended_animals	LIST	Stores the animals recommended to the users	Multiple animals will need to be displayed in ranked order so a list is a suitable data type	Ordered from lowest to highest compatibility

selected_page	STRING	Stores the page selected from navigation menu	This allows the program to choose which page or section should be selected	Animals or questionnaire
notification_message	STRING	Stores confirmation or status feedback	The user will need clear feedback after completing actions such as submitting questionnaire	'Questionnaire submitted successfully'

Animal Viewing Page

Animal Viewing Page

Animal Viewing	Allows users to browse available animals using clearly presented animal cards. Each card will show information such as the animal name, image, species, breed and age. Users will be able to select a card to view a more detailed animal profile.	Animal records will be retrieved from the animals table. Search and filter selections will be used to change the database query or filter the retrieved records. Selecting an animal will use its unique animal ID to load the correct profile.
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Interface Design

Company logo and name

Home

Browse animals

Questionnaire

My account

← Log out

Find your next companion

SEARCH AND FILTER

Search

Search by animal name or breed

Species

All species

Size

Any size

Clear Filters

Available animals



Luna

Dog • Labrador cross

3 years • Female • Large

Friendly and energetic. Luna enjoys long walks, training games and being around people.

View profile



Milo

Cat • Domestic shorthair

5 years • Male • Medium

Calm and affectionate. Milo would enjoy a quiet home with cosy places to relax.

View profile



Poppy

Rabbit • Mini lop

2 years • Female • Small

Gentle and curious. Poppy needs daily enrichment and space to explore safely.

View profile

Consistent branding and permanent navigation - Reusing these elements gives the application a recognisable identity as it matches the other pages. The main sections are in a predictable sidebar so users don't need to remember page routes or repeatedly return home. Text labels are used rather than relying on icons alone, improving clarity for different users.

Current page feedback - browse animals uses the filled pink active state while the other navigation controls are outlined. This immediate feedback confirms the users current location.

Secure logout placement - logout is separated from ordinary navigation and placed at the bottom of the sidebar. This reduces accidental selection and gives the user an obvious way to clear the authenticated session before returning to the login page.

Filtering - The species and size use controlled choices that align with the database fields and clear filters will restore the default view, when the selected criteria return too few results.

Page purpose is clear - The largest heading states the task before controls are shown. The position and size of the title are larger than the rest of the text so there is a clear visual hierarchy,

Consistent profile cards - every animal uses the same image area, information order, spacing and action position. This makes it easier for the user to navigate the different animals especially as they don't have to learn a different layout for each animal and makes coding the website easier.

Essential information first - Name, species, breed, age, sex, size and a short description provide enough information for an initial decision without overcrowding the page. The values should be read directly from the selected SQLite animal record.

Clear route to full details - View profile is the only strongly emphasised action on each card and has a large click target. Selecting it should store animal id and opening the matching animal profile preventing the wrong record from being displayed.

Flexible search - The search field allows records to be found by animal name and breed. The implementation should have validation so users can find a record effectively.

Company logo and name

Home

Browse animals

Questionnaire

My account

Log out

Find your next companion

Search by animal name or breed

Species: All species

Size: Any size

Clear Filters

Available animals



Luna
Dog • Labrador cross
3 years • Female • Large
Friendly and energetic, Luna enjoys long walks, training games and being around people.

View profile



Milo
Cat • Domestic shorthair
1 year • Male • Medium
Calm and affectionate, Milo would enjoy a quiet home with lots of places to relax.

View profile



Poppy
Rabbit • Mini Lop
2 years • Female • Small
Gentle and curious, Poppy needs daily enrichment and space to explore safely.

View profile

Company logo and name

Home

Browse animals

Questionnaire

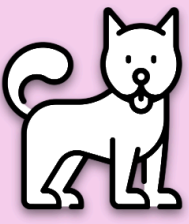
My account

Log out

Back to all animals

Meet Luna

Dog • Labrador cross



Luna description

Luna description in more depth
...

Age
3 years

Sex
Female

Size
Large

What [Name] Needs from a Home

Activity
High

Home Type
House

Garden
Required

Children
Yes

Time alone
4 hours

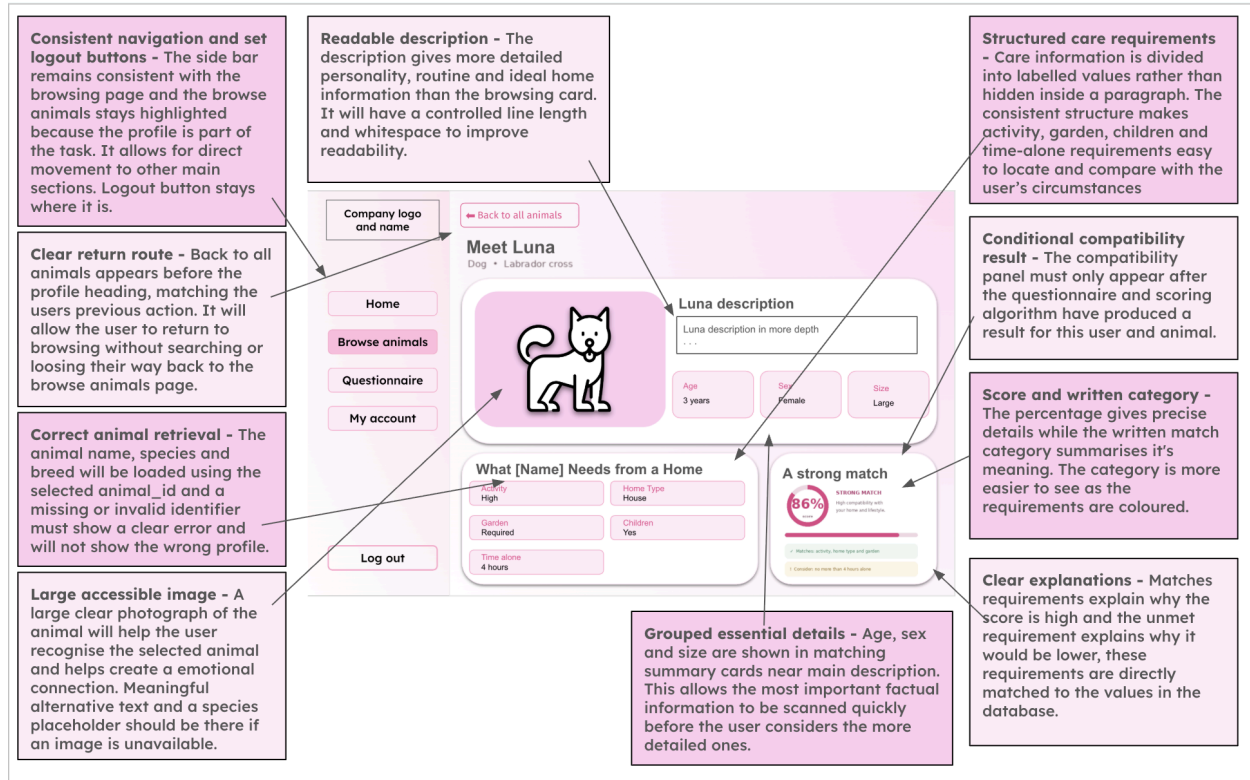
A strong match

86% score

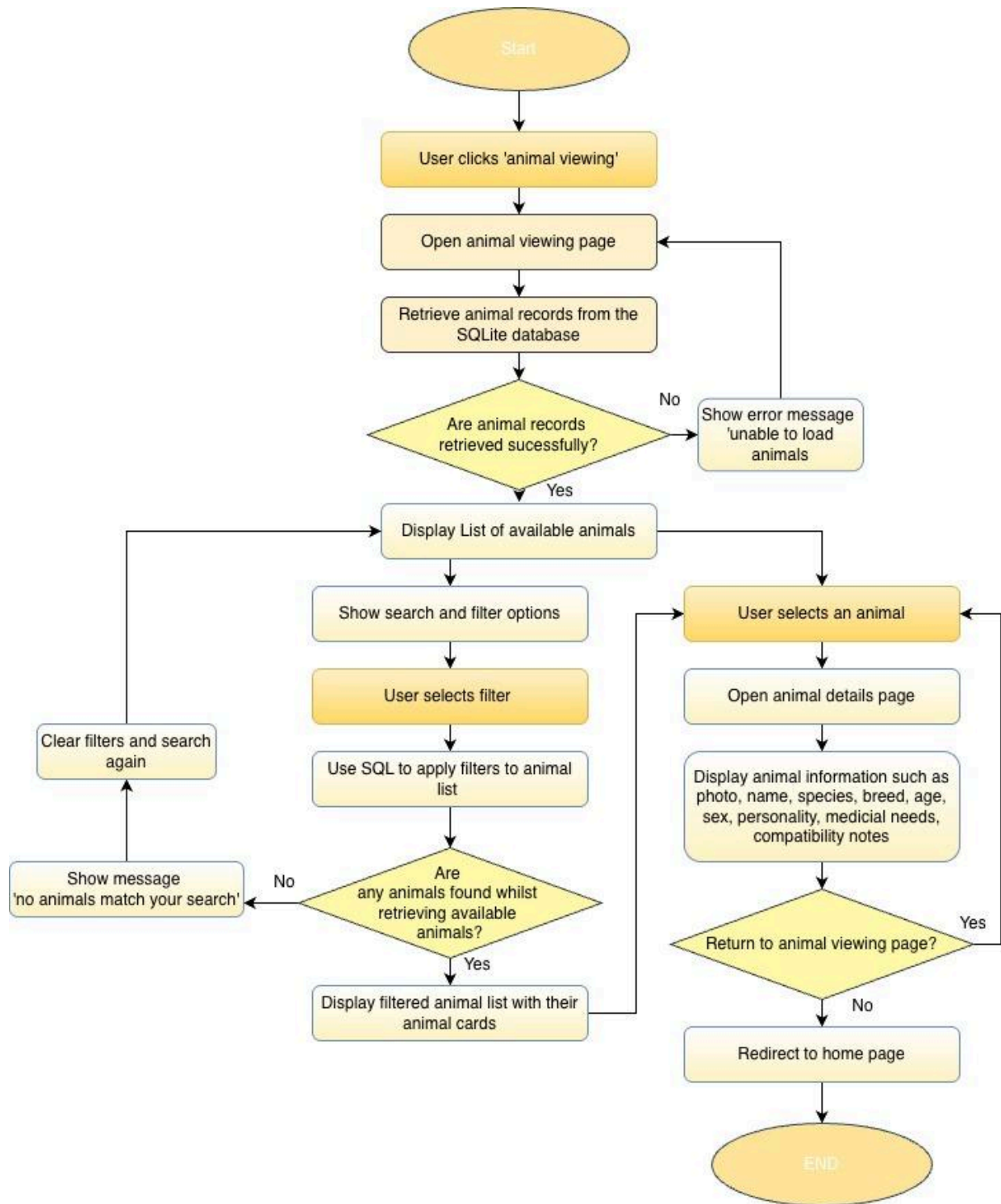
STRONG MATCH
High compatibility with your home and lifestyle.

Matches: activity, home type and garden

Consider: no more than 4 hours alone



Algorithm (Flow chart, pseudocode)



Pseudocode

Variables

Variable Name	Data Type	Purpose	Justification	E.g or validation
search_term	String	Stores text entered into the search bar	A string allows users to search using names, breeds or other information	Leading and trailing spaces should be removed
species_filter	String	Filters animals by species	This is required to reduce the displayed records to animals matching the users chosen species	'Dog', 'cat', 'rabbit' are the specified filters
size_filter	String	Filters animals by size	Size may affect whether an animal is suitable for the users home so it is suitable for narrowing the results	'Small', 'medium', large'
sort_option	String	Determines the order of results	Users may wish to order the results by suitability or animal information. The value determines which sorting operation is applied.	'Compatibility'
animal_results	list	Holds the records returned from the database	A list is suitable because a search may return multiple animal records. It can be iterated through to generate animal cards on the interface	May be empty is no matching results
selected_animal_id	integer	Identifies the animal whose profile is opened	Using the unique id ensures the correct animal is retrieved	Must match an existing record

			even when animals have the same name	
compatibility_score	float	Displays or sorts by compatibility	A float is a more suitable data type than an integer as it allows the result to include decimal values, which will produce a more precise % than an integer.	Between 0 and 100
no_results	boolean	Feedback displayed if there is any animals which meet the filters	Without feedback an empty page could appear to be an error, so the message informs the user that they should change the filters	'No animals match your search'

Validation

1. Search input

- Length check: the text will not exceed 20 characters
- Whitespace check: leading and trailing spaces will be removed
- Empty input handling: an empty search will be treated as a request to display all available animals rather than as an error

2. Filter inputs

Drop down menus or selection controls will be used for:

- Species
 - Dog
 - Cat
 - Rabbit
- Size
 - Small
 - Medium

- Large

Compatibility algorithms

Compatibility algorithm and questionnaire

Questionnaire and Recommendations	Will use a form in order to allow users to answer questions about their home, lifestyle, experience and preferences. The questionnaire will have questions such as the housing type, access to garden, animal loneliness times, children, pets, experience, activity labels, what species and size preferred. These questions would not appear all at one time to not overwhelm the user and the input type would be drop downs or radio buttons. After submitting the questionnaire, the system will display compatibility scores almost immediately and recommended animals. It will also explain requirements the user does not meet.	Responses will be validated and compared with the requirements of each available animal. A weighted compatibility algorithm will calculate a percentage score for every animal. The animals will then be sorted from highest to lowest score so the top recommendations can be displayed. The questionnaire responses and match results may also be stored within the database with the users ID
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Interface Design

Company logo and name

Home

Browse animals

Questionnaire

My account

Log out

25% through

Question 2 of 8

Choose one answer then continue

How much indoor space is available for the animal?

☐ Limited space
One main room

☒ Moderate Space
Two or three rooms

☐ Plenty of space
Most of the home

Back

Next question

Consistent navigation and set logout buttons - The side bar remains consistent, and the questionnaire section is now highlighted, which is especially useful as this page is a multistep section. It allows for direct movement to other main sections and easier familiarisation with the website. Logout button stays where it is.

Clear question position - The large 'question 2 of 8' at the top of the page, tells the user exactly where they are in the sequence. Showing both the current step and total number makes task feel finite and allows the remaining effort to be estimated.

Visible progress feedback - the progress bar shows that the system has recorded movement through the questionnaire.

Safe backwards - Back is visually next to next question, previously selected answers should remain in session state allowing the user to review or change a response without losing later questionnaire data

Company logo and name

Home

Browse animals

Questionnaire

My account

Log out

25% through

Question 2 of 8

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Back

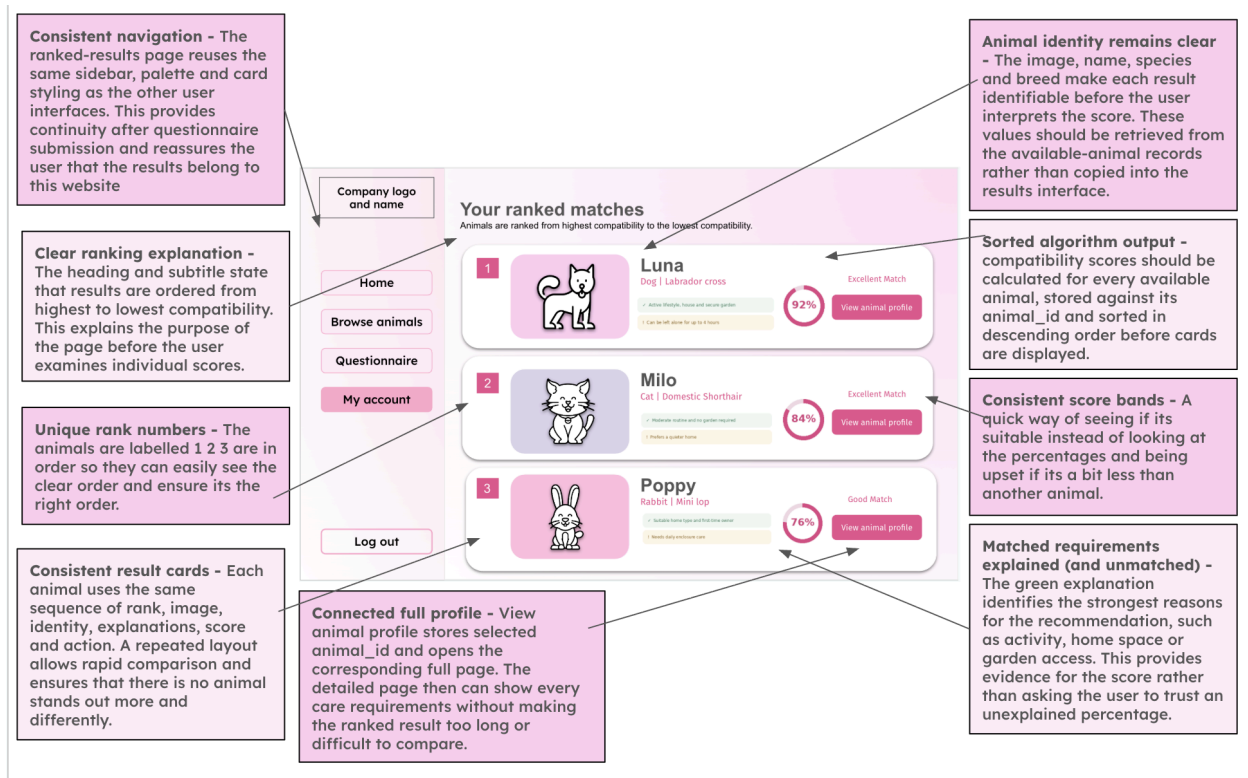
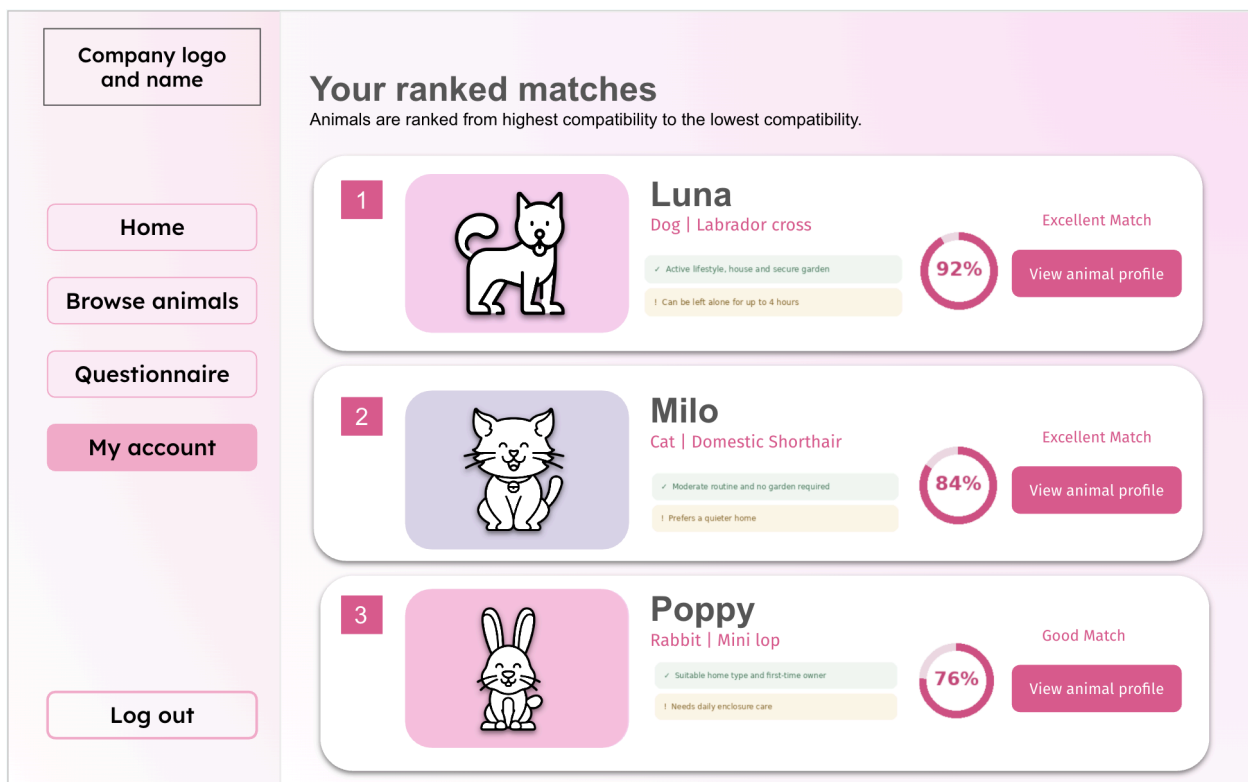
Next question

Clear options in their own boxes - Limited, moderate and plenty of space use a single choice control because only one answer should be stored. The room examples make the categories more objective and reduce different interpretation of vague words.

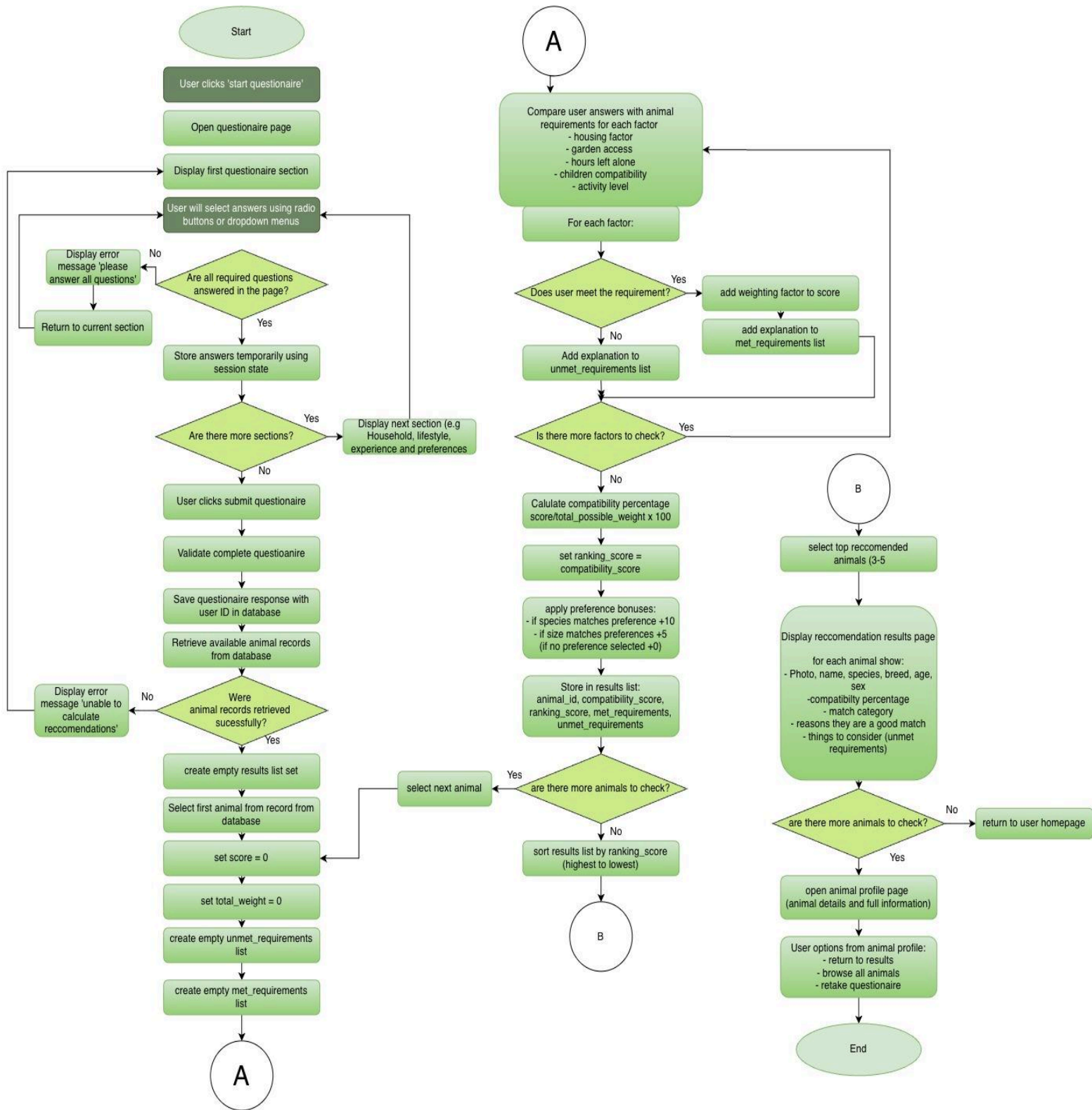
One question at a time - Displaying one question at a time removed unrelated actions and buttons and reduces cognitive load. The large central card focuses the attention on current decision and is easier to use. This links to the examples of competitors and their designs and the stakeholder survey and client with the website being easy to use.

Clear selected state - The chosen card uses a filled radio marker so selection will not change on a subtle colour change so it is very clear to the user what they have chosen as an option.

Validated progression- Next question should only progress if an answer is selected. A error message is shown if the user attempt to continue without responding.



Algorithm (Flow chart, pseudocode)



Pseudocode

Validation - questionnaire

1. Housing type or housing level

- Presence check: The user must select an answer
- Lookup check: The answer must be one of the permitted housing options
- A radio button or drop-down will prevent unsupported values

1 = Flat or limited space

2 = Small or medium house

3 = Large house

2. Garden access

- Presence check: Yes or No must be selected
- Boolean check: The stored result must be 1 or 0

3. Hours the animal will be left alone

- Presence check : a value must be provided
- Type check : a whole number
- Range check : between 0 and 24 hours

4. Children in the home

- Presence check: Yes or No must be selected
- Boolean check: The stored result must be 1 or 0

5. Preferred activity level

- Presence check: an option must be selected
- Boolean check: The stored result must be 1, 2 or 3

After

6. Preferred species

- Presence check: The user must select an answer
- Lookup check: The answer must be a dog, cat, rabbit or no preference

7. Preferred size

- Presence check: The user must select an answer
- Lookup check: answer must be small, medium, large or no preference

Validation is necessary within this compatibility algorithm to ensure that the recommendations produced are reliable and that invalid data does not cause incorrect results.

Questionnaire Response table

Field	Data type	Key/validation
response_id	INTEGER	Primary key, auto increment
user_id	INTEGER	Foreign key linked to Users.userid
housing_level	INTEGER	Required range 1-3
garden_access	BOOLEAN	Required 0 or 1
hours_left_alone	INTEGER	Required valid range
has_children	BOOLEAN	Required 0 or 1
preferred_activity_level	INTEGER	Required range 1 - 3
preffered_species	TEXT	Required dog, cat, rabbit or no preference
preferred_size	TEXT	Required small, medium, large or no preference
completed_at	DATE	required

Match Results Table

Field	Data type	Key/validation
match_id	INTEGER	Primary key autoincrement
response_id	INTEGER	Foreign key linked to QuestionnaireReponses.response_id
animal_id	INTEGER	Foreign key linked to animals.animal_id
compatibility_score	REAL	Required range 0 to 100
match_category	TEXT	Required strong, good, possible or low
unmet_requirements	TEXT	optional

Calculated_at	DATE	required
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Adoption Questionnaire Variables

Variable Name	Data Type	Purpose	Justification	Validation or e.g
housing_level	integer	Stores amount of living space available	The available space may affect whether or not the home is suitable for different sizes and exercise needs	Required range 1-3
garden_access	boolean	Records whether user has access to a garden	Some animals may require regular access to an outdoor space. A boolean value is suitable as the answer has two possible values	0 or 1
hours_left_alone	integer	Stores how long the animal would normally be left alone	Time availability can be numerically compared with the animals required attention and exercise time	Required valid range such as 0-24
has_children	boolean	Records whether children live in the household	Some animals may not be suitable for homes with children making this an important safety factor	0 or 1
preferred_activity_level	integer	Stores the users preferred animal activity	Matching activity level helps ensure the adopter can meet	range 1 - 3

		level	the animals exercise requirements	
preferred_species	string	Stores the users preferred species	This ensures that the recommendations remain relevant to the type of animal the user is interested in adopting	Required 'dog', 'cat', 'rabbit' or 'no preference'
preferred_size	string	Stores the users preferred size	The user may have practical or personal reasons for preferring a particular size.	'small', 'medium', 'large' or 'no preference'

Compatibility scoring algorithm variable names

Variable Name	Data Type	Purpose	Justification	Validation or e.g
animal_id	integer	Stores the unique identifier of the animal being assessed	It ensures that each calculated score is linked to the correct animal, even when animals have the same name	Must match an existing record in the animals table e.g 12
maximum_score	integer	Stores the highest possible number of points that can be awarded by the compatibility algorithm	This is required so that the user's raw score can be converted into a percentage. An integer is suitable because the algorithm awards whole number points for each matching factor.	Must be greater than 0

total_score	integer	Stores the total compatibility points awarded for the animal currently being assessed	The value begins at zero and increases whenever the user's answers from the questionnaire match an animal's requirements. It must be reset before the algorithm assesses the next animal.	Begins at 0 and cannot exceed maximum_score
animal_scores	list	Stores the ID and compatibility score for all available animals	This will allow the results to be sorted from highest to lowest compatibility. A list is suitable because the algorithm processes multiple animals	[{"animal_id": 3, "score": 92.5}, {"animal_id": 7, "score": 78.0}]
compatibility_score	float	Stores the calculated compatibility percentage	A percentage is easier for users to understand and compare than a raw points total, a float will allow decimal results to be stored.	Must be between 0-100 and can have decimals
match_category	string	Stores an understandable category for the calculated score	A category makes the numerical result easier for the user to interpret. It is selected according to predefined percentage ranges.	strong, good, possible or low
unmet_requirements	list	Stores explanations of requirements the user did not meet	There may be several reasons why an animal is not fully suitable so a list will allow multiple explanations to be stored and displayed.	Example: "Requires a secure garden", "Not suitable with cats". It may be empty when all essential requirements are met.

Questionnaire Plan

Section 1: House environment

- Amount of living space
- Garden access?

Section 2:

- Children in the home?

Section 3:

- Preferred activity level / time available for exercise and care
- Hours the animal would be left alone

Section 4:

- Preferred species
- Preferred animal size

Section 5:

- Submit

Where it links

- Amount of living space → minimum housing level
- Garden access → garden required
- Children in the home → child compatible
- Preferred activity level → exercise need
- Hours left alone → maximum hours left alone
- Preferred species → animal species
- Preferred size → animal size

First Matching part

Factor	Suggested weight (can be changed)	Justification
Hours Left alone	20	Strongly will affect whether the adopter can meet the animals needs

Child compatibility	15	Important for safety and suitability
Housing space	15	Ensures the home provides enough room
Activity	12	Matches exercise needs with users lifestyle and what they can provide
Garden access	10	Important for animals requiring outdoor access, but can even use a park so not as important factor

- Calculate the suitability score
- Remove animals with low suitability
- Prioritise animals matching users preferred species
- Prioritise animals matching users preferred size
- Sort remaining animals by suitability score
- Show top 3 or 5 (depending how many animals are in the system)

Factor	Suggested weight
Species	10
Size	5

However, these weights will not add to the compatibility percentage as their preferences do not reflect animal suitability. It will influence the order in which suitable animals are displayed. This will prevent an unsuitable animal from receiving a misleading high compatibility percentage as it matches the users preferences, but still ensuring preferred animals are presented as users top choices as this will make them more inclined to adopt. If the user picks 'no preference' no animal will receive a bonus.

Development Test plan

Development Test Plan